

## MBA Fastrack 2025 (CAT + OMETs)

## QUANTITATIVE APTITUDE

DPP: 5

## Mixture &amp; Allegation

- Q1** An ore of copper initially contains 60% gangue. After a refining process that removes only the gangue, the resultant refined copper contains 20% gangue. How much of the original ore is needed to produce 50 kg of the refined copper?  
 (A) 62.5 kg (B) 75 kg  
 (C) 80 kg (D) 100 kg
- Q2** A shopkeeper buys two lamps for Rs. 1200. He sells the first lamp at  $\frac{3}{5}$  of its cost price and the second lamp at  $\frac{5}{3}$  of its cost price. If his total profit from both the transactions is Rs. 80, what is the cost price of the lesser expensive lamp (in Rs.)?  
 (A) 75 (B) 425  
 (C) 525 (D) 675
- Q3** Four litres of a  $m\%$  sugar solution is mixed with seven litres of a  $n\%$  sugar solution to form a 60% sugar solution. If  $m > n$  and both  $m$  and  $n$  are positive integers, how many pairs of  $(m, n)$  satisfy this condition?
- Q4** A chemist has 30 litres of a chemical-water solution containing 40% chemical-X. He combines this solution with ' $c$ ' litres of another solution containing 70% chemical-X, creating a new mixture that is 60% chemical-X. Afterwards, the chemist adds another ' $c$ ' litres of a solution containing 90% chemical-X to this mixture. What is the percentage of chemical-X in the final solution?  
 (A) 50 (B) 62  
 (C) 70 (D) 72
- Q5** A milk vendor has two cans of milk. The first can contains milk with 40% fat, and the second can contains milk with 25% fat. How many liters from each can respectively should be mixed together to get 30 liters of milk containing 32% fat?  
 (A) 14 and 16 (B) 10 and 20  
 (C) 15 and 15 (D) 12 and 18
- Q6** Two containers, P and Q, hold mango juice and orange juice in the ratio 8:3 and 3:1, respectively. Container P has 77 litres of the mixture, while container Q contains 64 litres. After mixing the entire contents of P and Q, how many litres of orange juice must be added to the mixture so that the ratio of mango juice to orange juice becomes 13 : 8?
- Q7** There are two paint buckets, each containing a mixture of red and blue paint. In the first bucket, red and blue paint are in a ratio of 7:5. Paint from the first bucket and the second bucket are mixed in a ratio of 4:1. Red and blue paint are in a ratio of 13:11 in the resulting mixture. In the second bucket, the ratio of red to blue paint is:  
 (A) 3 : 5 (B) 3 : 8  
 (C) 5 : 3 (D) 8 : 3
- Q8** A water-detergent cleaning solution can be prepared by mixing 200 ml of a 70% concentrated detergent with ' $x$ ' ml of a 40% concentrated detergent. The same solution (in terms of concentration) can also be prepared by combining ' $x$ ' ml of an 80% concentrated detergent with 100 ml of a 40% concentrated detergent. Find the final concentration (in %) of the water-detergent cleaning solution.



- Q9** Fruit juices are blended with different proportions of apple, orange, and mango. Juice 1 contains **a%** apple and **b%** orange, with the rest being mango. Juice 2 contains **a%** orange and **b%** mango, with the rest being apple. Juice 3 contains **b%** apple and **a%** mango, with the rest being orange. When these three juices are mixed in the ratio 4:1:3 respectively, the final blend has 19% apple. Find the smallest possible value of 'b' if 'a' is an integer.
- (A) 1 (B) 2  
(C) 0.50 (D) 0.25
- Q10** A chemist has a total of 'x' number of bottles, each having the same cost price, of three kinds of solutions with different concentrations such that he earns a profit of 8%, 9%, and 10% respectively on selling them. For the first two solutions combined, he earns a profit of  $8\frac{5}{9}\%$ , while the overall profit on all three solutions is  $9\frac{5}{6}\%$ . Find the lowest possible integer value of x.
- Q11** Raman buys milk at Rs. 70/litre and mixes it with water which was available free of cost. Later, he sold the mixture at the same rate at which he purchased the milk. Doing this he gains 10%, what is the quantity of water mixed by him per litre of milk?
- (A) 0.5 litres  
(B) 0.25 litres  
(C) 0.1 litre  
(D) 0.01 litre
- Q12** There are two alloys. Alloy P is made up of **50%** gold, **20%** silver and **30%** bronze while Alloy Q contains only gold and silver. An alloy is formed by entirely mixing these two alloys. In the resultant alloy, concentration of gold is **50%**, silver is **40%** and bronze is **10%**. Then what is the concentration of gold and silver in alloy Q?

- (A) **60%** and **40%** respectively  
(B) **40%** and **60%** respectively  
(C) **55%** and **45%** respectively  
(D) None of these

- Q13** In a mixture of 'x' litres, the quantity of milk is **20%** more than the quantity of water. In another mixture of 180 litres, the quantity of water is **50%** more than the quantity of milk. After mixing both the mixtures, the ratio of milk to water becomes **10 : 11**, find the value of x.
- (A) 198 (B) 192  
(C) 200 (D) 208
- Q14** An alloy x composed of gold and silver is mixed with 25 kg of pure gold, and the resulting alloy y contains  $83.\overline{33}\%$  gold. If same quantity of alloy x is instead combined with 18 kg of another alloy containing 70% gold, the resulting alloy z contains 80% gold. What is the quantity of alloy x?
- (A) 133 kg (B) 136 kg  
(C) 156 kg (D) 179 kg
- Q15** A shopkeeper purchases two different types of flour. The cheaper variety is priced at Rs. 45 per kg, while the more expensive variety costs Rs. 55 per kg. When combined, the price of the mixture is a whole number in Rs. per kg. Furthermore, the quantity of the cheaper flour exceeds that of the expensive one by 6 kg. What is the maximum possible total weight of the flour mixture in kilograms?
- Q16** Sarah purchased varying amounts of three types of flour – X, Y, and Z – stored in three separate containers. When the containers holding flour X and Y are completely mixed, the average price of the mixture is Rs. 35/kg. When the containers with flour Y and Z are entirely combined, the average price is Rs. 49/kg. Likewise, when the containers with flour Z and X are completely mixed, the average



price of the mixture is Rs. 67/kg. Which of the following values cannot be the average cost per kg of flour when all three containers are mixed together?

- (A) Rs 42 (B) Rs 44  
(C) Rs 45 (D) Rs 49

**Q17** Two containers, X and Y, hold mixtures of alcohol and water. Container X has a composition of 36% alcohol, while Container Y contains 72% alcohol. When the mixtures from these two containers are combined in the ratio  $m:n$ , the resulting mixture has an alcohol-to-water ratio of  $n:m$ . If the resultant mixture has integral amounts of alcohol and water (in liters), then which of the following can be the quantity of alcohol in the resultant mixture?

- (A) 30 l (B) 39 l  
(C) 48 l (D) 54 l

**Q18** A farmer purchases 140 kg of corn and 150 kg of wheat, with corn being 20% cheaper per kg than wheat. He sells 40 kg of corn at a 10% profit and 50 kg of wheat at a 20% profit. Mixing the remaining corn and wheat, the farmer sells the mixture to a local factory at ₹344.96 per kg and achieves an overall profit of 68%. The cost price of wheat, in rupees per kg, is:

**Q19** Container X holds 4 liters of a 42% sugar solution. Container Y holds 6 liters of a 74% sugar solution. The solution in Container Y is completely poured into Container X along with some quantity of  $z\%$  sugar solution. If the final mixture in Container X is 12 liters of a 55% sugar solution, what is the value of  $z$ ?

- (A) 34 (B) 42  
(C) 24 (D) 28

**Q20** A woman mixes two liquids with varying Alcohol concentration. Liquid A has 35% alcohol while Liquid B has 50% alcohol. Find the ratio in which these the liquids A and B

should be mixed in order to obtain a new liquid containing 42% alcohol.

- (A) 8:7 (B) 7:8  
(C) 3:4 (D) 4:3

**Q21** A container holds 20 litres of milk and 10 litres of water mixture. 6 litres of the mixture is removed from the container, and 6 litres of water is added to it. This process is repeated two more times. What is the ratio of the remaining quantity of milk to its initial quantity?

- (A) 11 : 20 (B) 8 : 27  
(C) 27 : 1000 (D) 64 : 125

**Q22** A tank contains a mixture of oil and vinegar in a ratio of 2:7. Then, 36 litres of the mixture is removed from the tank, and 93 litres of oil is added. The ratio of oil to vinegar in the tank becomes 7:9. What was the initial quantity of oil in the tank (in litres)?

**Q23** A vessel contains 64 litres of a mixture of petrol and ethanol in a ratio of 7: 5. 8 litres of mixture is taken out and replaced by 14 litres of petrol. Again, 6 litres of mixture is taken out and replaced by 8 litres of ethanol. What is the ratio of petrol and ethanol in the resulting mixture?

- (A) 16 : 17 (B) 15 : 11  
(C) 16 : 11 (D) 15 : 17

**Q24** A 200-litre mixture of milk and water has 25% milk in it. One-fifth of the mixture is removed and replaced with an equal quantity of water and this process is repeated several times till the milk concentration comes down to 16% of the new solution. Find the number of replacements performed.

**Q25** A tank containing pure juice has 25% of its volume removed, after which 6 litres of water is added to it. Then, 20% of the new mixture is removed, and 12 litres of water is added. Finally, 80% of the resulting mixture is



removed, and 24 litres of water is added. If the final ratio of pure juice to water is 3:4, find the initial volume of pure juice in the tank (in litres).

- Q26** Three different paints have a mixture of red and blue colours in the ratios of 4:5, 5:6, and 9:7. 18 litres from the first paint and 22 litres from the second are taken. How much of the third paint (in litres) should be taken to achieve a final colour mixture with a red-to-blue ratio of 1:1?
- (A) 30 (B) 32  
(C) 34 (D) 36
- Q27** A tank contains 800 litres of acetic acid. First, 40 litres of acetic acid is removed and replaced with water. Then, 16% of the resulting solution is replaced with water. Finally,  $\frac{1}{6}$ th of the solution is replaced with water. Find the amount of acetic acid (in litres) remaining in the solution at the end.
- Q28** A tank labelled T is completely filled with **2 litres** of pure honey. A group of workers pour  $\frac{1}{3}$ rd of the contents of Tank T into a large empty jar. Then, they refill Tank T to its brim with pure water without any spillage and pour **50%** of the mixture from Tank T into the jar. Finally, they refill Tank T again to its brim with pure water and completely empty its contents into the jar. What is the concentration (in %) of honey in the mixture now contained in the jar?
- (A)  $57\frac{2}{11}$  (B)  $54\frac{2}{11}$   
(C)  $54\frac{6}{11}$  (D)  $57\frac{6}{11}$
- Q29** A vendor sells fruit-water juice prepared by mixing two types of juices, one containing 90% concentrate and the other containing 80% concentrate, in a ratio of 3:1. Wanting to maximize profits, he further mixes this solution with another juice containing 60% concentrate in a ratio of 8:3. Find the profit

percentage of the vendor if he marks up the price of pure concentrated juice by 40% and sells the final solution at that price. (Assume the cost of water to be negligible).

- Q30** A **container** has 600 cc of **orange** juice, and a pitcher contains 600 cc of apple juice. First, 120 cc of orange juice is transferred to the pitcher and mixed thoroughly with the apple juice. Then, 120 cc of this mixed juice is transferred back from the pitcher **to the container**. Finally, 60 cc from the container is taken out and thrown away. Now, the ratio of the remaining orange juice in the container to apple juice in the pitcher is:
- (A) 1 : 1 (B) 1 : 5  
(C) 10 : 1 (D) 9 : 10



# Answer Key

Q1 D  
Q2 C  
Q3 5  
Q4 D  
Q5 A  
Q6 27  
Q7 A  
Q8 60  
Q9 C  
Q10 78  
Q11 C  
Q12 D  
Q13 A  
Q14 D  
Q15 30

Q16 A  
Q17 D  
Q18 200  
Q19 C  
Q20 A  
Q21 D  
Q22 62  
Q23 C  
Q24 2  
Q25 171  
Q26 B  
Q27 532  
Q28 C  
Q29 75  
Q30 D



# Hints & Solutions

Note: scan the QR code to watch video solution

## Q1 Text Solution:

Let the original ore be = 100x  
 Gangue = 60x and Copper = 40x  
 As the refining process removes only the Gangue part, the quantity of copper remains the same i.e. 40x  
 This 40x forms 80% of the refined copper, thus refined copper would be

$$\frac{40}{80} \times 100 = 50x$$

Therefore, from 100x of the original ore, we get 50x of the refined copper.  
 In order to have 50 kg of the refined copper. 100 kg of the original ore is required.

### Video Solution:



## Q2 Text Solution:

Given that, he sells the first lamp at  $\frac{3}{5}$  of its cost price, so he sells at  $\frac{3}{5} \times 100 = 60\%$  of the cost price. Therefore, he makes a loss of  $(100 - 60) = 40\%$  or a profit of  $-40\%$ .  
 Again, he sells the second lamp at  $\frac{5}{3}$  of its cost price, so he sells at  $\frac{5}{3} \times 100 = \frac{500}{3}\%$  of its cost price. Thus, he makes a profit of  $(\frac{500}{3} - 100) = \frac{200}{3}\%$ .  
 Now, it is also given that, overall in both two transactions, he earns a profit of Rs. 80. i.e.,

$$\frac{80}{1200} \times 100 = \frac{20}{3}\% \text{ profit.}$$

So, by the method of Alligations, we have

|                   |                    |
|-------------------|--------------------|
| <u>First Lamp</u> | <u>Second Lamp</u> |
| - 40%             | $\frac{200}{3}\%$  |
|                   | $\frac{20}{3}\%$   |

For *smooth calculations*, multiplying each term by 3 and rewriting,

| <u>First Lamp</u> | <u>Second Lamp</u> |
|-------------------|--------------------|
| - 120%            | 200%               |
|                   | 20%                |
| 180               | : 140              |
| = 9 : 7           |                    |

So, the total cost price =  $(9 + 7) = 16$  units = Rs. 1200 (given)  
 1 unit = Rs. 75  
 Hence, the cost price of the less expensive Lamp = Rs.  $(7 \times 75) = \text{Rs. } 525$ .

### Video Solution:



## Q3 Text Solution:

Note that, according to the condition, since,  $m > n$ , so  $m > 60 > n \dots (i)$   
 By the Alligation method, we have

|                 |          |
|-----------------|----------|
| m%              | n%       |
|                 | 60%      |
| 60 - n          | : m - 60 |
| = 4 : 7 (Given) |          |

Since, 4 units of m% sugar is mixed with 7 units of n% sugar to give 60% sugar, so

$$\frac{60-n}{m-60} = \frac{4}{7}$$

$$\Rightarrow 60 - n = \frac{4(m-60)}{7}$$

$$\Rightarrow n = 60 - \frac{4(m-60)}{7}$$

To get positive integer values of n, the value of  $(m - 60)$  must be a multiple of 7. Also, by (i),  $m > 60$ .  
 Therefore, m can be 67, 74, 81, 88, and 95.





So, for these 5 positive integer values of  $m$ , we will get 5 positive integer values of  $n$ .

#### Video Solution:



#### Q4 Text Solution:

Using the Alligation method for first the two chemical-water solutions:

| <u>Chemical-X in 1st solution</u> | <u>Chemical-X in 2nd solution</u> |
|-----------------------------------|-----------------------------------|
|-----------------------------------|-----------------------------------|

|           |     |
|-----------|-----|
| 40%       |     |
| 70%       |     |
|           | 60% |
| 10        | 20  |
| $= 1 : 2$ |     |

So, if the 1st solution is of 30 litres, then the 2nd solution must be of 60 litres.

Therefore,  $c = 60$  litres

So, the total volume of the mixture =  $(60 + 30)$  = 90 litres and it is of concentration 60% chemical-X.

Let the final mixture be of  $x\%$  chemical-X. Then, again, using the Alligation method, we have

| <u>Chemical-X in 3rd solution</u> | <u>Chemical-X in 4th solution</u> |
|-----------------------------------|-----------------------------------|
|-----------------------------------|-----------------------------------|

|                                                                  |          |
|------------------------------------------------------------------|----------|
| 60%                                                              | 90%      |
|                                                                  | $x\%$    |
| $90 - x$                                                         | $x - 60$ |
| $= 90 \text{ litres} : 60 \text{ litres}$ [As obtained earlier.] |          |
| Therefore, we have                                               |          |
| $\frac{90-x}{x-60} = \frac{90}{60}$                              |          |
| $\frac{90-x}{x-60} = \frac{3}{2}$                                |          |
| $180 - 2x = 3x - 180$                                            |          |
| $5x = 360$                                                       |          |
| $x = 72\%$                                                       |          |

Hence, the concentration of chemical-X in the final solution is 72%.

#### Video Solution:



#### Q5 Text Solution:

Using Alligation method, we have,

| First Can                                                                    | Second Can   |
|------------------------------------------------------------------------------|--------------|
| 40                                                                           | 25           |
|                                                                              | 32           |
| $7(32 - 25)$                                                                 | $8(40 - 32)$ |
| Ratio of mixing is $7 : 8$ ( $7 + 8 = 15$ parts)                             |              |
| Total quantity required is 30 litres                                         |              |
| Therefore, the quantity that needs to be taken from each of the cans will be |              |
| $7 \times 2 = 14$ and $8 \times 2 = 16$                                      |              |

#### Video Solution:



#### Q6 Text Solution:

Container P contains, mango juice : orange juice =  $8 : 3$

Container Q contains, mango juice : orange juice =  $3 : 1$

77 litres of the mixture in container P contains mango juice =  $77 \times \frac{8}{11} = 56$  litres, and orange juice =  $77 - 56 = 21$  litres  
 64 litres of the mixture in container Q contains mango juice =  $64 \times \frac{3}{4} = 48$  litres, and orange juice =  $64 - 48 = 16$  litres  
 Therefore, after mixing both these mixtures, the ratio becomes, mango juice : orange juice =  $(56 + 48) : (21 + 16) = 104 : 37$



Now, some amount of orange juice is added to the mixture. So, there was no change of the mango juice in ratio.

And, the final ratio of the mixture is, mango juice : orange juice = 13 : 8 = 104 : 64

So, the total amount of orange juice added was = (64 - 37) = 27 litres

#### Video Solution:



#### Q7 Text Solution:

The concentration of the red paint in the first bucket =  $\frac{7}{12}$

The concentration of the red paint in the mixture of two paints =  $\frac{13}{24}$

As,  $\frac{7}{12} > \frac{13}{24}$ , therefore, the concentration of the red paint in the second bucket is less than  $\frac{13}{24}$ .

Therefore, by the Alligations method, we get

$$\begin{array}{ccc} \frac{7}{12} & & \frac{13}{24} \\ & \times & \\ \left(\frac{13}{24} - x\right) & : & \left(\frac{7}{12} - \frac{13}{24}\right) \end{array}$$

Now, given that,

$$\left(\frac{13}{24} - x\right) : \left(\frac{7}{12} - \frac{13}{24}\right) = 4 : 1$$

$$\left(\frac{13}{24} - x\right) = \frac{4}{1} \times \frac{1}{24} = \frac{1}{6}$$

$$x = \frac{13}{24} - \frac{1}{6} = \frac{3}{8}$$

Thus, in the second bucket, the required ratio was,

red paint : blue paint = 3 : (8 - 3) = 3 : 5

#### Video Solution:



#### Q8 Text Solution:

The concentration of all the mixtures remains the same.

The concentration of solution 1 is =

$$\frac{200 \times 70\% + x \times 40\%}{200 + x}$$

Similarly, the concentration of solution 2 is =

$$\frac{x \times 80\% + 100 \times 40\%}{x + 100}$$

Since, the two concentrations are the same, so we have

$$\frac{200 \times 70\% + x \times 40\%}{200 + x} = \frac{x \times 80\% + 100 \times 40\%}{x + 100}$$

$$(x + 100)(140 + 0.4x)$$

$$= (200 + x)(0.8x + 40)$$

$$14000 + 40x + 140x + 0.4x^2 = 160x + 0.8x^2 + 8000 + 40x$$

$$0.4x^2 + 20x - 6000 = 0$$

$$x^2 + 50x - 15000 = 0$$

$$(x + 150)(x - 100) = 0$$

$x = 100$ , since, the volume cannot be -ve.

Thus, the required concentration

$$= \frac{0.8x + 0.4 \times 100}{100 + x} = \frac{80 + 40}{200} = 60\%$$

#### Video Solution:



#### Q9 Text Solution:

Juice-1: a% apple, b% orange, (100 - a - b)% mango.

Juice-2: a% orange, b% mango, (100 - a - b)% apple.





Juice-3:  $b\%$  apple,  $a\%$  mango,  $(100 - a - b)\%$  orange.

Since, the final blend has 19% apple, and three juices are mixed in the ratio of 4: 1: 3, so by the weighted-average rule, we have

$$\frac{a \times 4 + 1 \times (100 - a - b) + 3 \times b}{4 + 1 + 3} = 19$$

$$4a + 100 - a - b + 3b = 152$$

$$3a + 2b = 52$$

$$b = \frac{52 - 3a}{2}$$

Since,  $a$  is a positive integer, and we need the smallest value of  $b$ , so when  $a = 17$ , we get the smallest value of  $b = \frac{52 - 3 \times 17}{2} = 0.5$

**Video Solution:**



**Q10 Text Solution:**

By using the method of Alligations for the profit% of the first two solutions, we get

Profit on 1st solution

Profit on 2nd solution

$$8\% \times 9 = 72\%$$

$$9\% \times 9 = 81\%$$

$$8\frac{5}{9}\% \times 9 = \frac{77}{9} \times 9\% = 77\%$$

$$= \frac{4}{5} :$$

i.e., ratio of bottles, 1st solution : 2nd solution  
= 4 : 5

So, total bottles in the first two solutions =  $(4 + 5) = 9$  units

Again, using the method of Alligations for the overall profit% from the first two solutions and the third solution, we get

Overall profit on the first two solutions  
profit on 3rd solution

$$8\frac{5}{9}\% \times 18 = 154\%$$

$$10\% \times 18 = 180\%$$

$$9\frac{5}{6}\% \times 18 = 177\%$$

$$= \frac{3}{23} :$$

i.e., ratio of bottles, 1st two solutions

combined : 3rd solution = 3 : 23 = 9 : 69

So, the ratio of bottles, 1st solution : 2nd

solution : 3rd solution = 4 : 5 : 69

Thus, the total number of bottles =  $4 + 5 + 69 = 78$  units, which is the lowest integer number of bottles in total.

**Video Solution:**



**Q11 Text Solution:**

Let Raman buy 10 liters of milk at Rs. 70/liter. Suppose he mixed  $x$  liter of water in it.

Now, the actual quantity in a liter of mixture  
=  $(10 + x)$

So, selling price at same rate =  $70(10 + x)$

Now, according to the question.

$$\frac{SP - CP}{CP} \times 100 = \text{Gain}$$

$$\frac{700 + 70x - 700}{700} \times 100 = 10$$

$$\text{or } 70x = 70$$

$$\text{or } x = 1$$

So, per liter of water mixed

$$= \frac{1}{10}$$

$$= 0.1 \text{ liter}$$

Ans. (c)



**Video Solution:****Q12 Text Solution:****Topic - Mixtures and Alligations**

In Alloy P,  $G : S : B = 5 : 2 : 3$

In Alloy Q,  $G : S = x : y$ ,

In resultant alloy,  $G : S : B = 5 : 4 : 1$

Since Bronze can be obtained from Alloy P only, so concentration of bronze in Alloy P and resultant alloy is constant.

$$G : S : B \text{ in resultant} = (5 : 4 : 1)3 = 15 : 12 : 3$$

In Alloy Q,

Ratio of Gold : Ratio of Silver

$$= (15 - 5) : (12 - 2) = 10 : 10 = 1 : 1$$

i.e., Concentration of Gold = Concentration of Silver = 50%.

Hence, option d is correct.

**Video Solution:****Q13 Text Solution:**

In a mixture of  $x$  litres,

$$\text{Ratio of milk and water} = 120 : 100 = 6 : 5$$

$$\text{Quantity of milk} = \left(\frac{x}{11}\right) \times 6$$

$$\text{Quantity of water} = \left(\frac{x}{11}\right) \times 5$$

In another mixture of 180 litres,

$$\text{Ratio of milk and water} = 100 : 150 = 2 : 3$$

$$\text{Quantity of milk} = \left(\frac{180}{5}\right) \times 2 = 72 \text{ litres}$$

$$\text{Quantity of water} = \left(\frac{180}{5}\right) \times 3 = 108 \text{ litres}$$

According to the question,

$$\left\{\left(\frac{x}{11}\right) \times 6 + 72\right\} : \left\{\left(\frac{x}{11}\right) \times 5 + 108\right\} = 10 : 11$$

$$11 \left\{\left(\frac{x}{11}\right) \times 6 + 72\right\}$$

$$= 10 \left\{\left(\frac{x}{11}\right) \times 5 + 108\right\}$$

$$6x + 792 = \frac{50x}{11} + 1080$$

$$6x - \frac{50x}{11} = 1080 - 792$$

$$\frac{16x}{11} = 288$$

$$x = 198$$

Hence, option a is correct.

**Video Solution:****Q14 Text Solution:**

Let the amount of gold and silver in alloy x be  $g$  kg and  $s$  kg, so

In y

$$\frac{g+25}{g+s+25} = \frac{83.33}{100}$$

$$\Rightarrow 6g + 150 = 5g + 5s + 125$$

$$\Rightarrow g = 5s - 25$$

In z

$$\frac{g+18(0.7)}{g+s+18} = \frac{80}{100}$$

$$\Rightarrow 5g + 63 = 4g + 4s + 72$$

$$\Rightarrow g = 4s + 9$$

So using the value of  $g$  in the previous equation

$$5s - 25 = 4s + 9$$

$$\Rightarrow s = 34$$

$$g = 4s + 9 = 145$$

$$\text{So weight of } x = 34 + 145 = 179 \text{ kg}$$

**Video Solution:**

**Q15 Text Solution:**

Let the cost of the mixture be Rs x, so

Quantity of cheaper : quantity of expensive =  
(55-x):(x-45)

Quantity of cheaper = quantity of expensive+6

Since quantity of cheaper is more, 55-x>x-45,  
so x<50

| x  | (55-x):(x-45)=a:b | Total  |
|----|-------------------|--------|
| 49 | 3:2               | 30 kg  |
| 48 | 7:3               | 15 kg  |
| 47 | 4:1               | 10 kg  |
| 46 | 9:1               | 7.5 kg |

So maximum weight of the mixture will be 30 kg.

**Video Solution:****Q16 Text Solution:**

Let the price of the three flours be Rs x, Rs y and Rs z respectively and let their quantities be a kg, b kg and c kg respectively. So

$$ax+by=35a+35b$$

$$by+cz=49b+49c$$

$$ax+cz=67a+67c$$

So adding the three equations

$$2(ax + by + cz) = 102a + 84b + 116c$$

$$\Rightarrow ax + by + cz = 51a + 42b + 58c$$

So to find the average price we must divide it by  $(a + b + c)$ , so

$$\frac{ax+by+cz}{a+b+c} = \frac{51a+42b+58c}{a+b+c}$$

$$\Rightarrow \frac{ax+by+cz}{a+b+c} = 42 + \left( \frac{9a+16c}{a+b+c} \right) = 58 - \left( \frac{7a+16b}{a+b+c} \right)$$

So the average cost per kg when all the three flours are mixed together should be between Rs 42 and Rs 58, excluding both. So option (a) is not possible.

**Video Solution:****Q17 Text Solution:**

Working on alcohol percentage

$$\frac{36 \times m + 72 \times n}{m+n} = \frac{n}{m+n} \times 100$$

$$\Rightarrow 36m + 72n = 100n$$

$$\Rightarrow 36m = 28n$$

$$\Rightarrow \frac{m}{n} = \frac{7}{9}$$

The ratio of alcohol to water in the resultant mixture is n:m, i.e., 9:7

So their quantities in resultant mixture can be 9x and 7x, which means the quantity of alcohol will be a multiple of 9, so 54l.

**Video Solution:****Q18 Text Solution:**

Let the cost price of the wheat per kg is ₹x.



Then, the cost price of corn per kg is  $= x \times 80\%$   
 $= ₹0.8x$

Now, the total cost price of 140 kg of corn and 150 kg of wheat  $= (150x + 140 \times 0.8x) = ₹262x$

Therefore, the selling price of 40 kg of corn  $= 110\% \times 40 \times 0.8x = ₹35.2x$

The selling price of 50 kg of wheat  $= 120\% \times 50 \times x = ₹60x$

Remaining corn  $= (140 - 40) = 100$  kg and  
 remaining wheat  $= (150 - 50) = 100$  kg.

The total selling price of 140 kg of corn and 150 kg of wheat after having an overall 68% profit  $= 262x \times 168\% = ₹440.16x$

Hence, the selling price of the remaining mixture  $= 440.16x - 35.2x - 60x = ₹344.96x$

Thus, we have

$$344.96x = 344.96 \times (100 + 100)$$

$$x = 200$$

Hence, the cost price of wheat per kg  $= ₹200$

#### Video Solution:



#### Q19 Text Solution:

If only the initial contents of container X and container Y would have been mixed and the resulting solution is  $m\%$  sugar solution, then

$$\frac{74-m}{m-42} = \frac{4}{6}$$

$$\Rightarrow 444 - 6m = 4m - 168$$

$$\Rightarrow m = 61.2$$

Since the final quantity of the solution is 12 l, so it means 2 l of  $z\%$  sugar solution was added, so

$$\frac{61.2-55}{55-z} = \frac{2}{10}$$

$$\Rightarrow 612 - 550 = 110 - 2z$$

$$\Rightarrow 2z = 48$$

$$\Rightarrow z = 24$$

#### Video Solution:



#### Q20 Text Solution:

Applying Alligation rule, we have,

| Liquid A | Liquid B |
|----------|----------|
| 35       | 50       |

42

$$8 (50 - 42)$$

$$7 (42 - 35)$$

Therefore, the ratio in which the liquids A and B must be mixed is 8 : 7

#### Video Solution:



#### Q21 Text Solution:

Let  $x$  = the initial amount of milk in the container = 20 litres

$y$  = the initial amount of water in the container = 10 litres

$a$  = The amount of mixture removed and replaced by water = 6 litres

$n$  = The number of times the same process is repeated = 3

So, according to the well-known formula, we have the required ratio

$$\begin{aligned} &= \left(1 - \frac{a}{x+y}\right)^n \\ &= \left(1 - \frac{6}{20+10}\right)^3 \\ &= \left(1 - \frac{6}{30}\right)^3 \\ &= \left(\frac{4}{5}\right)^3 \\ &= 64 : 125 \end{aligned}$$

#### Video Solution:



**Q22 Text Solution:**

Initial ratio, oil : vinegar = 2 : 7

36 litres of the mixture was taken out. Then, in these 36 litres of the mixture, the ratio of oil to vinegar is the same, 2 : 7.

So, in these 36 litres of the **taken-out mixture**, the amount of oil =  $36 \times \frac{2}{2+7} = 8$  litres

After removing 36 litres of the mixture, the remaining mixture still has the same oil-to-vinegar ratio intact.

Now, again, 93 litres of oil is added. So, there should be no change of vinegar in the ratio.

The final ratio, oil : vinegar = 7 : 9

**Revised ratios,**

Initial ratio, oil : vinegar = 18 : 63

Final ratio, oil : vinegar = 49 : 63

So,  $(49 - 18) = 31$  unit ratio = 93 litres

i.e., 1 unit ratio = 3 litres

i.e., the initial quantity of oil in the drum was =

$(18 \times 3) + 8 = 54 + 8 = 62$  litres

**Video Solution:****Q23 Text Solution:**

In the initial amount of 64 litres of mixture, petrol : ethanol = 7 : 5

After taking out 8 litres of mixture, in  $(64 - 8) = 56$  litres of mixture, the ratio of petrol and ethanol must be the same as before, i.e., 7 : 5

Therefore, now, 56 litres = 12 units ratio

So, 14 litres = 3 units ratio.

i.e., after adding 14 litres of **petrol** to the mixture, new ratio, petrol : ethanol =  $(7 + 3) : 5 = 10 : 5 = 2 : 1$

Again, it is said, 6 litres of mixture is taken out and replaced by 8 litres of **ethanol**.

So, when 6 litres of mixture is taken out, in the remaining mixture of  $(56 + 14 - 6) = 64$  litres, petrol : ethanol = 2 : 1

i.e., now, 64 litres = 3 units ratio

8 litres =  $\frac{3}{8}$  units ratio.

Thus, in the final mixture, petrol : ethanol = 2 :

$(1 + \frac{3}{8}) = 16 : 11$

**Video Solution:****Q24 Text Solution:**

**Final Concentration = Initial Concentration**

$\times (1 - \text{Fraction of Quantity Removed and Replaced})^n$   
where 'n' represents the number of times the process is carried out.

$$16 = 25 \times \left(1 - \frac{1}{5}\right)^n$$

$$\left(\frac{4}{5}\right)^n = \frac{16}{25} = \left(\frac{4}{5}\right)^2$$

$$n = 2$$

**Video Solution:****Q25 Text Solution:**

Let's solve the problem in a tabular format:



|                                                                                                     | Pure Juice in the tank                       | Water in the tank                                                 |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------|
| <b>Initially</b>                                                                                    | x litres (say)                               | 0                                                                 |
| <b>After the 1st step:</b><br>(After removing 25% of the volume and adding 6L of water)             | 75% of x = $\frac{3x}{4}$ litres             | 6 litres                                                          |
| <b>After the 2nd step:</b><br>(After removing 20% of the new mixture and adding 12L of water)       | 80% of $\frac{3x}{4} = \frac{3x}{5}$ litres  | $6 \times \frac{80}{100} + 12 = \frac{84}{5}$ litres              |
| <b>After the 3rd step:</b><br>(After removing 80% of the resulting mixture and adding 24L of water) | 20% of $\frac{3x}{5} = \frac{3x}{25}$ litres | $\frac{84}{5} \times \frac{20}{100} + 24 = \frac{684}{25}$ litres |

So, finally, the required ratio, pure juice : water  
 $= \frac{3x}{25} : \frac{684}{25} = 3x : 684 = 3 : 4$  (Given)  
Therefore,  $3x = 3$  i.e.,  $x = 1$  unit.  
Also, 4 units = 684, and so, 1 unit =  $x = 171$  litres

Video Solution:



Q26 Text Solution:

In the 1st paint, Red : Blue = 4 : 5  
In 18 litres of the first paint, the quantity of Red colour =  $18 \times \frac{4}{9} = 8$  litres

and the quantity of Blue color =  $18 - 8 = 10$  litres

In the 2nd paint, Red : Blue = 5 : 6  
In 22 litres of the 2nd paint, the quantity of Red color =  $22 \times \frac{5}{11} = 10$  litres  
and the quantity of Blue color =  $22 - 10 = 12$  litres

So, till now, we have  $(8 + 10) = 18$  litres of Red and  $(10 + 12) = 22$  litres of Blue color. But, we need to make the ratio 1:1.  
Therefore, we need 4 litres of Red color more from the third paint.

In the third paint, Red : Blue = 9 : 7  
It means the quantity of Red color is 2 litres more than Blue color if we take 16 litres of color from the third paint.  
So, to get 4 litres more Red color than Blue, we need to take  $\frac{16}{2} \times 4 = 32$  litres of the third paint.

Video Solution:



Q27 Text Solution:

When for the first time, 40 litres is removed from 800 litres of acetic acid, the left-over ratio of acetic acid  
 $= \frac{800-40}{800} = \frac{760}{800}$   
Again, when for the second time, 16% of the resulting solution is replaced with water, the left-over ratio of acetic acid  
 $= \frac{100-16}{100} = \frac{84}{100}$   
For the third time, when  $\frac{1}{6}$ th of the solution is replaced with water, then the left-over ratio of acetic acid





$$= 1 - \frac{1}{6} = \frac{5}{6}$$

Thus, in the end, the amount of acetic acid remains in the final mixture =

$$800 \times \frac{760}{800} \times \frac{84}{100} \times \frac{5}{6} = 532 \text{ litres.}$$

**Video Solution:**



#### Q28 Text Solution:

Let's solve this in a **tabular format**:

| Step                                                      | Tank T<br>(Honey)    | Tank T<br>(Water)    | Jar<br>(Honey)       | Jar<br>(Water)       |
|-----------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|
| <b>Initial State</b>                                      | 2 liters             | 0 liters             | 0 liters             | 0 liters             |
| <b>Step 1:</b>                                            |                      |                      |                      |                      |
| <b>Pour <math>\frac{1}{3}</math>rd from Tank T to Jar</b> | $\frac{4}{3}$ liters | 0 liters             | $\frac{2}{3}$ liters | 0 liters             |
| <b>Step 2:</b>                                            |                      |                      |                      |                      |
| <b>Refill T with Water</b>                                | $\frac{4}{3}$ liters | $\frac{2}{3}$ liters | $\frac{2}{3}$ liters | 0 liters             |
| <b>Step 3:</b>                                            |                      |                      |                      |                      |
| <b>Pour 50% of T to Jar</b>                               | $\frac{2}{3}$ liters | $\frac{1}{3}$ liters | $\frac{4}{3}$ liters | $\frac{1}{3}$ liters |
| <b>Step 4:</b>                                            |                      |                      |                      |                      |
| <b>Refill T with Water</b>                                | $\frac{2}{3}$ liters | $\frac{4}{3}$ liters | $\frac{4}{3}$ liters | $\frac{1}{3}$ liters |
| <b>Step 5:</b>                                            |                      |                      |                      |                      |
| <b>Pour All of T to Jar</b>                               | 0 liters             | 0 liters             | 2 liters             | $\frac{5}{3}$ liters |
| <b>Final State</b>                                        | 0 liters             | 0 liters             | 2 liters             | $\frac{5}{3}$ liters |

Thus,

$$\text{total Volume in Jar} = 2 + \frac{5}{3} = \frac{11}{3} \text{ liters.}$$

#### Concentration of Honey in the Jar

$$= \left( \frac{\text{Honey Volume}}{\text{Total Volume}} \right) \times 100 = \left( \frac{2}{\frac{11}{3}} \right) \times 100$$

$$= \frac{600}{11} = 54\frac{6}{11} \%.$$

**Video Solution:**



#### Q29 Text Solution:

We are given that, the first solution is prepared by mixing 90% and 80% concentrated juice in a ratio of 3:1.

Assume 90% concentrated juice to be 30 litres having 27 litres of juice and 3 litres of water and

80% concentrated juice to be 10 litres having 8 litres of juice and 2 litres of water.

So, the solution will have **35** litres of juice and **5** litres of water.

Now this solution is mixed with 60% concentrated juice in a ratio of 8:3.

If the existing solution is of 40 litres, the 60% concentrated juice will be of 15 litres having **9** litres of juice and **6** litres of water.

So, the final solution will have  $(35 + 9) = 44$  litres of juice and  $(5 + 6) = 11$  litres of water.

Assuming the cost price of pure juice to be Rs. 100.

So, after marking up the selling price of pure juice is Rs. 140.

He will receive  $55 \times 140 = \text{Rs. } 7700$  by selling the mixture which has a cost price of Rs. 100.

The profit % of the vendor will be =

$$\frac{7700 - 4400}{4400} \times 100 = 75\%. \text{ [Since the cost of water is negligible.]}$$

**Video Solution:**



**Q30 Text Solution:**

120 cc of orange juice is transferred to the pitcher from the container.

So, now the **pitcher** has 600 cc of apple juice and 120 cc of orange juice.

and the container has  $(600 - 120) = 480$  cc of orange juice left.

So, the ratio of orange juice to apple juice in the **pitcher** =  $120 : 600 = 1 : 5$

Now, 120 cc of this mixed juice is collected **from the pitcher**. So, in this 120 cc mixture, it has  $120 \times \frac{5}{1+5} = 100$  cc apple juice and  $120 \times \frac{1}{1+5} = 20$  cc orange juice.

Since, this 120 cc of the mixture is transferred back to the container, so now, **in the**

**container**, apple juice present = **100** cc  
and orange juice present =  $480 + 20 = 500$  cc.

When 60 cc from the container is taken out, then the remaining apple juice in the container =  $100 - 60 \times \frac{1}{6} = 90$  cc and the remaining orange juice in the container =  $500 - 60 \times \frac{5}{6} = 450$  cc.

Now, in the **pitcher**, orange juice remains =  $120 - 20 = 100$  cc  
and apple juice remains =  $600 - 100 = 500$  cc  
So, the required ratio, orange juice in the container : apple juice in the pitcher =  $450 : 500 = 9 : 10$

**Video Solution:**
[Android App](#)
[iOS App](#)
[PW Website](#)